7.8 DeepLabv3

- denote by *output_stride* the ratio of input image spatial resolution to final output resolution
- for the CNNs deployed for the task of image classification, the final feature maps (before fully connected layers or global pooling) are 32 times smaller than the input image dimension, and thus *output_stride*$=32$
- if one would like to quadruple the spatial density of computed feature responses in the CNNs (i.e., *output_stride*$=8$), the strides of the last two pooling or convolutional layers that decrease resolution are set to 1 to avoid signal decimation
- then, all subsequent convolutional layers are replaced with dilated convolutional layers having rate $r = 2$

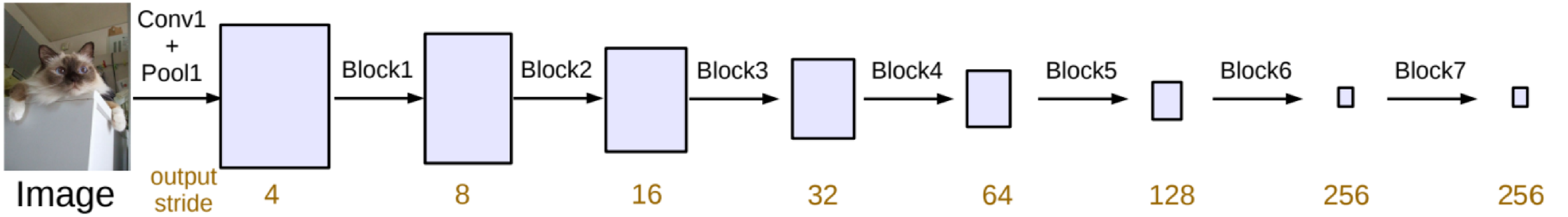
7.8 DeepLabv3

- first, modules with dilated convolution laid out in cascade are designed in the DeepLabv3 model
- to be concrete, several copies of the last ResNet block, denoted as block4 in Figure 16, are duplicated and arranged in cascade
- there are three convolutions in those blocks, and the last convolution contains stride 2 except the one in the last block, similar to the original ResNet
- the motivation behind this model is that the introduced striding makes it easy to capture long range information in the deeper blocks
- for example, the whole image feature could be summarized in the last small resolution feature map, as illustrated in Figure 16(a)
- however, it was discovered that the consecutive striding is harmful for semantic segmentation since detail information is decimated

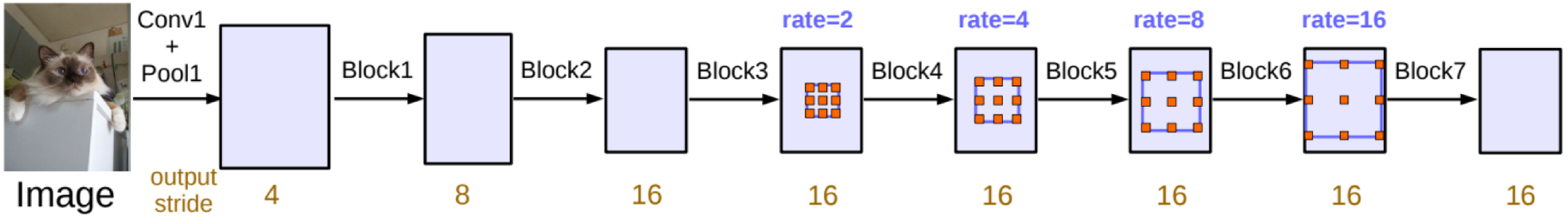
7.8 DeepLabv3

- thus, dilated convolution with rates determined by the desired output stride value are applied, as shown in Figure 16(b), where *output_stride*$=16$
- in this proposed model, cascaded ResNet blocks up to block7 (i.e., extra block5, block6, block7 as replicas of block4) are used, which has *output_stride*$=256$ if no dilated convolution is applied
- motivated by multi-grid methods which employ a hierarchy of grids of different sizes, different dilation rates within block4 to block7 in the DeepLabv3 model are adopted
- in particular, we define as *Multi_Grid*$=(r_1, r_2, r_3)$ the unit rates for the three convolutional layers within block4 to block7
- the final dilation rate for the convolutional layer is equal to the multiplication of the unit rate and the corresponding block rate
- for example, when *output_stride*$=16$ and *Multi_Grid*$=(1,2,4)$, the three convolutions will have *rates* $= 2 \cdot (1,2,4) = (2,4,8)$ in block4, respectively

7.8 DeepLabv3



(a) Going deeper without atrous convolution.



(b) Going deeper with atrous convolution. Atrous convolution with $rate > 1$ is applied after block3 when $output_stride = 16$.

Figure 16: Cascaded modules without and with dilated convolution.

7.8 DeepLabv3

- we revisit the Atrous Spatial Pyramid Pooling (ASPP) proposed in DeepLab, where four parallel dilated convolutions with different dilation rates are applied on top of the feature map
- ASPP is inspired by the success of spatial pyramid pooling, which showed that it is effective to resample features at different scales for accurately and efficiently classifying regions of an arbitrary scale
- different from DeepLab, batch normalization is included within ASPP
- ASPP with different dilation rates effectively captures multi-scale information
- however, it was discovered that, as the sampling rate becomes larger, the number of valid filter weights (i.e., the weights that are applied to the valid feature region, instead of padded zeros) becomes smaller

7.8 DeepLabv3

- to overcome this problem and incorporate global context information to the model, DeepLabv3 adopts image-level features
- specifically, it applies global average pooling on the last feature map of the model, feed the resulting image-level features to a 1×1 convolution with 256 filters (and batch normalization), and then bilinearly upsample the feature to the desired spatial dimension
- in the end, the improved ASPP consists of:
  1. one 1×1 convolution and three 3×3 convolutions with *rates* $= (6, 12, 18)$ when *output_stride*$= 16$ (all with 256 filters and batch normalization), and
  2. the image-level features, as shown in Figure 17
- the resulting features from all the branches are then concatenated and pass through another 1×1 convolution (also with 256 filters and batch normalization) before the final 1×1 convolution which generates the final logits

7.8 DeepLabv3

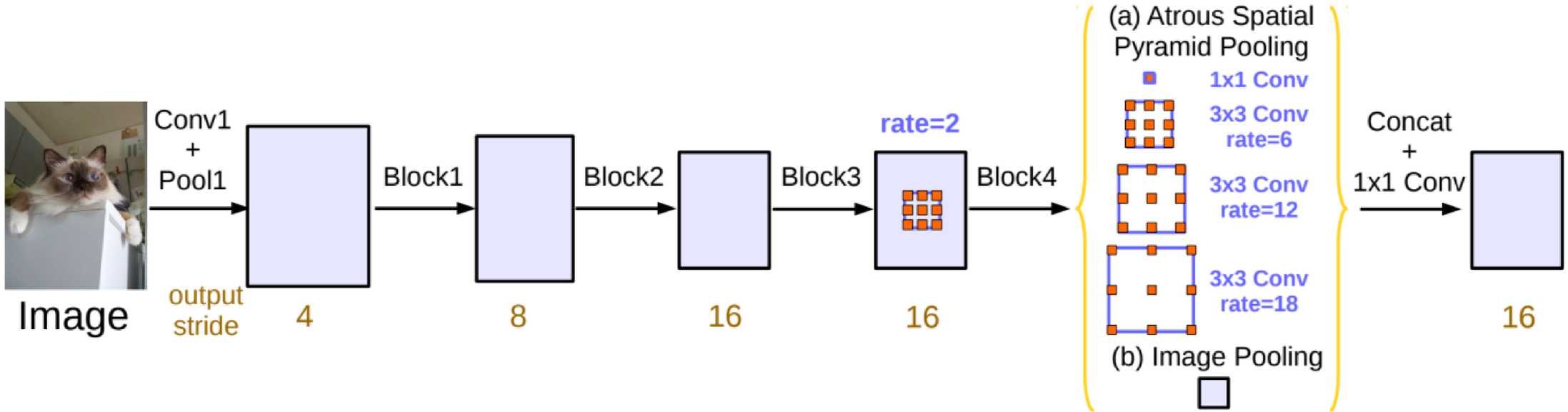


Figure 17: Parallel modules with dilated convolution (ASPP), augmented with image-level features.